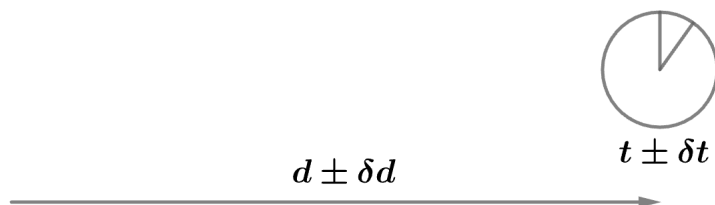


2018B F=ma Exam: Problem 19

Kevin S. Huang



Theory:

$$v = \frac{d}{t}$$

We have

$$d = 75.0 \pm 2.0 \text{ cm}$$

$$t = 2.15 \pm 0.10 \text{ ms}$$

Recall when two uncorrelated quantities are multiplied, the relative (fractional) uncertainties add in quadrature. Also recall the relative uncertainty gets multiplied by $|n|$ when the quantity is raised to the n th power. Thus,

$$v = \frac{d}{t} = \frac{75.0 \text{ cm}}{2.15 \text{ ms}} = 349 \text{ m/s}$$

$$\frac{\delta v}{v} = \sqrt{\left(\frac{\delta d}{d}\right)^2 + \left(\frac{-\delta t}{t}\right)^2}$$

$$\delta v = 349 \sqrt{\left(\frac{2.0}{75.0}\right)^2 + \left(\frac{0.10}{2.15}\right)^2} = 19 \text{ m/s}$$

so the answer is E.